MATH 152
Mrs. Bonny Tighe

EXAM IA

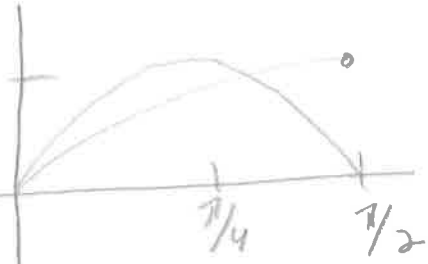
6.1-7.5
50 points
PART ONE

Name Solutions

TA_____ Wed 10/2/2019

1. (11 points) Find the area of the region bounded the curves $y = \sin 2x$, *and* $y = \sin x$ on the interval $0 \le x \le \frac{\pi}{2}$. Graph.

$$\int_0^{\pi/3} (\sin 2x - \sin x)\,dx \quad \int_{\pi/3}^{\pi/2} (\sin x - \sin 2x)\,dx$$



$$-\frac{1}{2}\cos 2x + \cos x \Big|_0^{\pi/3} + -\cos x + \frac{1}{2}\cos 2x \Big|_{\pi/3}^{\pi/2}$$

$$\left(-\frac{1}{2}\cos \tfrac{2\pi}{3} + \cos \tfrac{2\pi}{3}\right) - \left(-\tfrac{1}{2}+1\right) + \left(-\cos \tfrac{\pi}{2} + \tfrac{1}{2}\cos \pi\right) - \left(-\cos \tfrac{\pi}{3} + \tfrac{1}{2}\cos \tfrac{2\pi}{3}\right)$$

$$-\frac{1}{2}\left(-\frac{1}{2}\right) + -\frac{1}{2} + \frac{1}{2} + \frac{1}{2}(-1) + \frac{1}{2} - \frac{1}{2}\left(-\frac{1}{2}\right)$$

$$+\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$2\sin x \cos x = \sin x$
$\sin x(2\cos x - 1) = 0$
$\sin x \quad \cos x = \frac{1}{2}$
$x = \pi/3$

2. (3 points each) True or false Circle the correct True or False

True False a) The functions $y = \sin^2 x$ and $y = \cos^2 x$ only vary by a constant.

True False b) $\int \frac{x^2+1}{x^2-9}\,dx = \int \frac{A}{x-3} + \frac{B}{x+3}\,dx$

True False c) $\int \sec^4 y \tan^3 y\, dy$, can be integrated by using $u = \sec y$ and also $u = \tan y$.

True False d) The area between two curves could have a solution of $-\ln\left(\frac{3}{4}\right)$

True False e) If $\tan\theta = \frac{(x+2)}{4}$, then $\sec\theta = \frac{\sqrt{(x+2)^2 - 16}}{4}$

True False f) To evaluate $\int \cot^2(w)\,dw$ you need to use u-substitution.

True False g) $\frac{1}{1-\sin x} = 1 - \csc x$.

True False h) $\int x^2 e^{x^3}\,dx$ has to be evaluated using parts.

True False i) The partial fraction decomposition, without finding the constants, for $\frac{1}{x^3(x+1)}$ would be $\frac{Ax^2 + Bx + C}{x^3} + \frac{D}{x+1}$

True False j) The formula for the average value of a function is $f_{avg} = \int_a^b f(x)dx$

3. (3 points each) Multiple Choice. Circle the correct letter

a) How many terms does the partial fraction decomposition of $\frac{1}{x^5 + x^3}$ have.

$x^3(x^2+1)$

A) 4

B) 6

C) 1

D) 5

E) Some other answer

b) What method should NOT be used to integrate $\int \frac{x}{x^2-4}dx$

A) Trigonometric substitution

B) Partial Fraction Decomposition

C) Integration by Parts

D) U-Substitution

c) If the average value of the function $f(x)=4x-9x^2$ on the interval $[0,b]$ is -1, what is b?

A) 3

B) 1

C) $1/3$

D) 0

E) Some other answer

d) After making a substitution for $\int \frac{x^2\,dx}{\sqrt{16-x^2}}$

A) $\int 4\sin^2\theta \;\; d\theta$

B) $\int \frac{4\sin^2\theta}{\cos\theta}d\theta$

C) $\int 16\sin^2\theta\, d\theta$

D) $\int \frac{1}{\sqrt{u}}du$, where $u=16-x^2$

E) Some other answer

e) Using the Cylindrical Shells method to find the volume of the solid of revolution for the function $y = 5x - x^2$ about the line $x = 5$, you would have:

A) $2\pi \int_0^4 (5-x)(5x - x^2 + 5)dx$

B) $2\pi \int_0^4 (x-5)(5x - x^2)dx$

C) $2\pi \int_0^4 (5-x)(5x - x^2)\,dx$

D) $2\pi \int_0^4 (x-5)(5x - x^2 - 5)\,dx$

E) Some other answer

f) On which of the following can you NOT use the method of partial fractions?

A) $\int_0^1 \frac{1}{x^2 - 16}dx$

B) $\int_0^1 \frac{1}{x^3 + 16}dx$

C) $\int_0^1 \frac{1}{x^2 + 16}dx$

D) $\int_0^1 \frac{1}{x^4 - 16}dx$

E) Some other answer

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PART TWO

NAME Solutions

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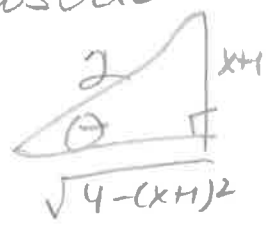
1. (10 points) Integrate $\int \frac{(x+1)^2}{\sqrt{3-x^2-2x}}\,dx$

$-(x^2+2x+1)+3+1$
$-(x+1)^2+4$

$x+1=2\sin\theta$
$dx=2\cos\theta\,d\theta$



$$\int \frac{(x+1)^2}{\sqrt{4-(x+1)^2}}\,dx = \int \frac{4\sin^2\theta\,(2\cos\theta\,d\theta)}{\sqrt{4-4\sin^2\theta}}$$

$$\int \frac{8\sin^2\theta\cos\theta\,d\theta}{\sqrt{4\cos^2\theta}} = 4\int \sin^2\theta\,d\theta$$

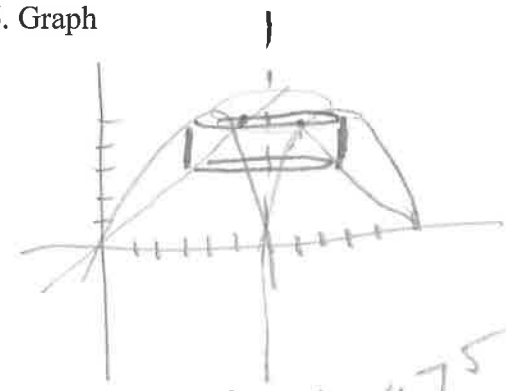
$$4\int\left(\tfrac{1}{2}-\tfrac{1}{2}\cos 2\theta\right)d\theta = 4\left[\tfrac{1}{2}\theta-\tfrac{1}{4}\sin 2\theta\right]+C$$

$$2\theta - 2\sin\theta\cos\theta + C =$$

$$2\sin^{-1}\left(\tfrac{x+1}{2}\right) - 2\left(\tfrac{x+1}{2}\right)\left(\frac{\sqrt{4-(x+1)^2}}{2}\right)+C$$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $y=6x-x^2$ *and* $y=x$ rotated about the line x = 6. Graph

$6x-x^2=x \qquad 5x-x^2=0 \qquad x=0,5$



$$2\pi\int_0^5 (6-x)\left[6x-x^2-x\right]dx$$

$$2\pi\int_0^5 (6-x)(5x-x^2)\,dx =$$

$$2\pi\int_0^5 (30x-6x^2-5x^2+x^3)\,dx$$

$$2\pi\int_0^5 (30x-11x^2+x^3)\,dx = 2\pi\left[15x^2-\tfrac{11}{3}x^3+\tfrac{1}{4}x^4\right]_0^5$$

$$2\pi\left[15(5^2)-\tfrac{11}{3}(5^3)+\tfrac{1}{4}(5^4)-0\right]$$

$$5^3(2\pi)\left[3-\tfrac{11}{3}+\tfrac{5}{4}\right] = 250\pi[$$

3. (10 points) Integrate: $\int \sin(\ln t)\, dt$

$u = \sin(\ln t)$ $\quad dv = dt$

$du = \cos(\ln t) \cdot \frac{1}{t}\, dt$ $\quad v = t$

4. (10 points) Integrate: $\int_3^6 \frac{x^3}{\sqrt{x^2-9}}\, dx$

$x = 3\sec\theta$ $\quad \sec\theta = x/3$

$dx = 3\sec\theta\tan\theta\, d\theta$ $\quad \sec\theta = 2$

$\theta = \pi/3$

$\sec\theta = 1$

$\theta = 0$

$$\int_0^{\pi/3} \frac{27\sec^3\theta\; 3\sec\theta\tan\theta\, d\theta}{\sqrt{9\sec^2\theta - 9}}$$

$$\int_0^{\pi/3} \frac{27\sec^4\theta\; 3\tan\theta\, d\theta}{\sqrt{9\tan^2\theta}} = \int_0^{\pi/3} 27\sec^4\theta\, d\theta$$

$$27\int_0^{\pi/3} \sec^2\theta(1+\tan^2\theta)\, d\theta = 27\int_0^{\pi/3} \sec^2\theta\, d\theta + 27\int_0^{\pi/3} \tan^2\theta\sec^2\theta\, d\theta$$

$$27\left[\tan\theta + \frac{1}{3}\tan^3\theta\right]_0^{\pi/3} =$$

$$27\left[\tan\pi/3 + \frac{1}{3}(\tan\pi/3)^3 - (0)\right]$$

$$27\left[\sqrt{3} + \frac{1}{3}(\sqrt{3})^3\right] = 27\left[\sqrt{3} + \sqrt{3}\right] = 54\sqrt{3}$$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\,dx$

6. (5 points) Integrate $\int \frac{e^x}{e^{2x}+1}\,dx$

$m = e^x$

$dm = e^x\,dy$

$\frac{dm}{m} = dx$

$\int \frac{1}{m^2+1}\,dm$

$\tan^{-1} m + C$

$= \tan^{-1}(e^x) + C$